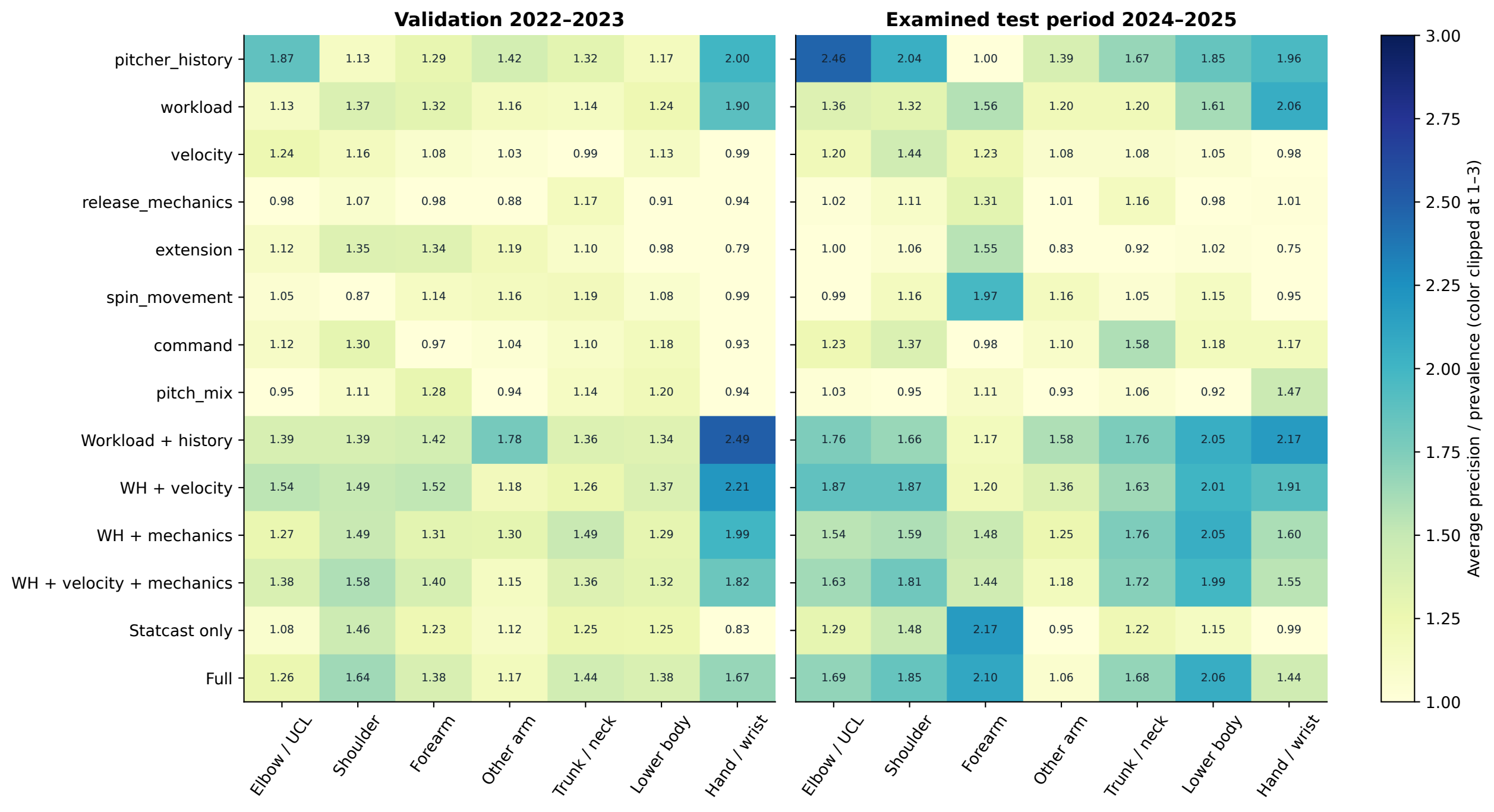

Validation 2022–2023
Examined test period 2024–2025
pitcher_history 1.87 1.13 1.29 1.42 1.32 1.17 2.00 2.46 2.04 1.00 1.39 1.67 1.85 1.96
workload 1.13 1.37 1.32 1.16 1.14 1.24 1.90 1.36 1.32 1.56 1.20 1.20 1.61 2.06
velocity 1.24 1.16 1.08 1.03 0.99 1.13 0.99 1.20 1.44 1.23 1.08 1.08 1.05 0.98
release_mechanics 0.98 1.07 0.98 0.88 1.17 0.91 0.94 1.02 1.11 1.31 1.01 1.16 0.98 1.01
extension 1.12 1.35 1.34 1.19 1.10 0.98 0.79 1.00 1.06 1.55 0.83 0.92 1.02 0.75
spin_movement 1.05 0.87 1.14 1.16 1.19 1.08 0.99 0.99 1.16 1.97 1.16 1.05 1.15 0.95
command 1.12 1.30 0.97 1.04 1.10 1.18 0.93 1.23 1.37 0.98 1.10 1.58 1.18 1.17
pitch_mix 0.95 1.11 1.28 0.94 1.14 1.20 0.94 1.03 0.95 1.11 0.93 1.06 0.92 1.47
Workload + history 1.39 1.39 1.42 1.78 1.36 1.34 2.49 1.76 1.66 1.17 1.58 1.76 2.05 2.17
WH + velocity 1.54 1.49 1.52 1.18 1.26 1.37 2.21 1.87 1.87 1.20 1.36 1.63 2.01 1.91
WH + mechanics 1.27 1.49 1.31 1.30 1.49 1.29 1.99 1.54 1.59 1.48 1.25 1.76 2.05 1.60
WH + velocity + mechanics 1.38 1.58 1.40 1.15 1.36 1.32 1.82 1.63 1.81 1.44 1.18 1.72 1.99 1.55
Statcast only 1.08 1.46 1.23 1.12 1.25 1.25 0.83 1.29 1.48 2.17 0.95 1.22 1.15 0.99
Full 1.26 1.64 1.38 1.17 1.44 1.38 1.67 1.69 1.85 2.10 1.06 1.68 2.06 1.44
Elbow / UCL
Shoulder
Forearm
Other arm
Trunk / neck
Lower body
Hand / wrist
Average precision / prevalence (color clipped at 1–3)
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00